ITC Hotel Revenue Optimization

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Objective

The primary objective of this project is to **analyze and optimize the revenue performance of ITC Hotels** through data-driven insights. By leveraging an **interactive, multi-page Power BI dashboard**, the goal is to enable stakeholders to make **real-time, informed decisions** that enhance profitability and customer satisfaction.

Specifically, the project aims to:

- **Assess overall financial performance** across hotels, room categories, and time periods.
- **Analyze booking behavior and occupancy trends** to identify demand patterns and key revenue drivers.
- **Evaluate room category performance** to optimize pricing and inventory allocation.
- **Monitor and reduce booking cancellations**, identifying root causes by channel, day type, and segment.
- **Develop actionable insights** that support strategic decisions around pricing, marketing, and distribution.
- **Empower leadership** with a unified, data-visual platform for continuous **revenue optimization and operational efficiency**.

Slicers

Finance

Room Occupancy

Room Performance

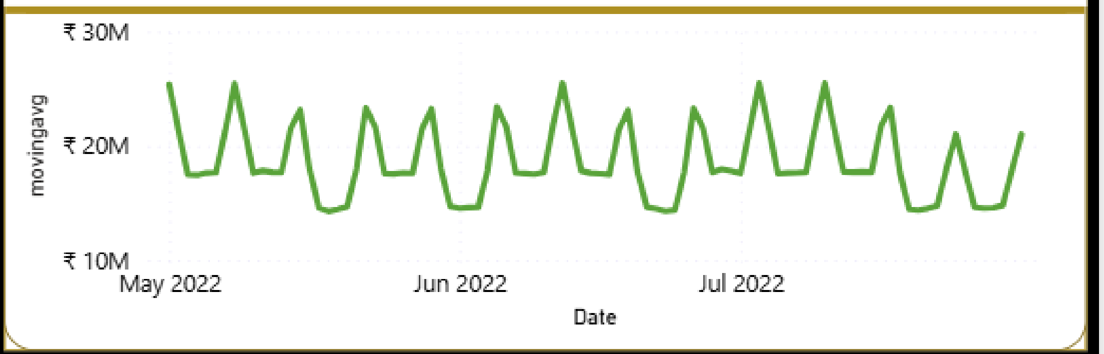
Cancellation

Financial Overview & Revenue Performance

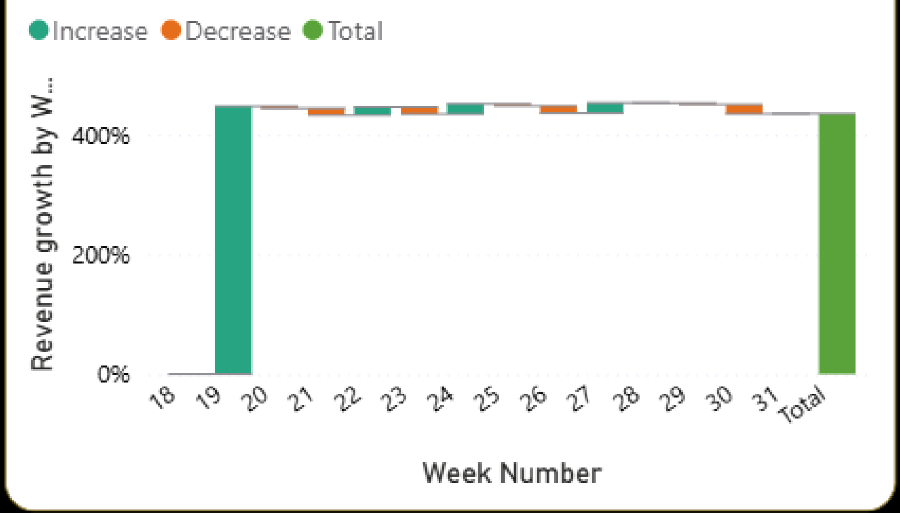
₹ 2.01bn	₹ 2.01bn	50.29%	1395.0...	₹ 7.35K	₹ 14.92K
Total Revenue	Cumulative Revenue	MoM_Growth	Wow_Growth	RevPAR	Average Daily Rate

property_id	Elite	Premium	Presidential	Standard
16558	₹ 1,72,88,460	₹ 84,01,680	₹ 57,56,240	₹ 1,08,05,34(
16559	₹ 3,89,93,580	₹ 4,17,09,840	₹ 3,70,22,260	₹ 2,09,72,90(
16560	₹ 2,15,40,960	₹ 1,72,97,280	₹ 94,69,600	₹ 1,58,30,36(
16561	₹ 1,87,29,900	₹ 2,26,19,520	₹ 1,68,40,460	₹ 1,03,78,55(
16562	₹ 2,74,25,160	₹ 1,59,80,160	₹ 87,08,840	₹ 1,41,78,71(
[illegible]	[illegible]	[illegible]	[illegible]	[illegible]
Total	**₹ 65,89,46,160**	**₹ 54,35,97,840**	**₹ 44,14,57,020**	**₹ 36,35,45,19!**

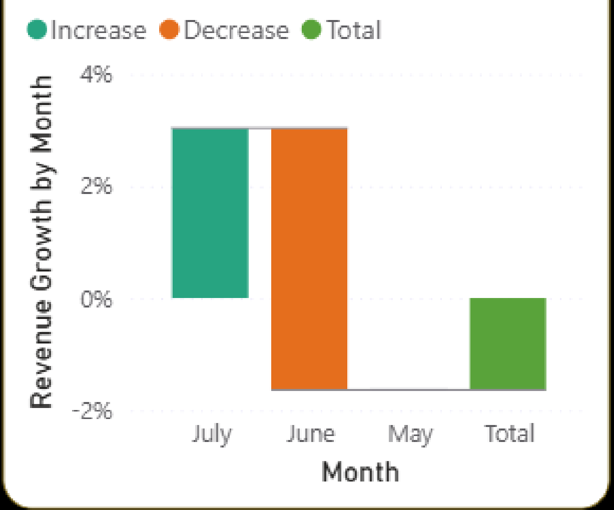
ADR | RevPAR | movingavg



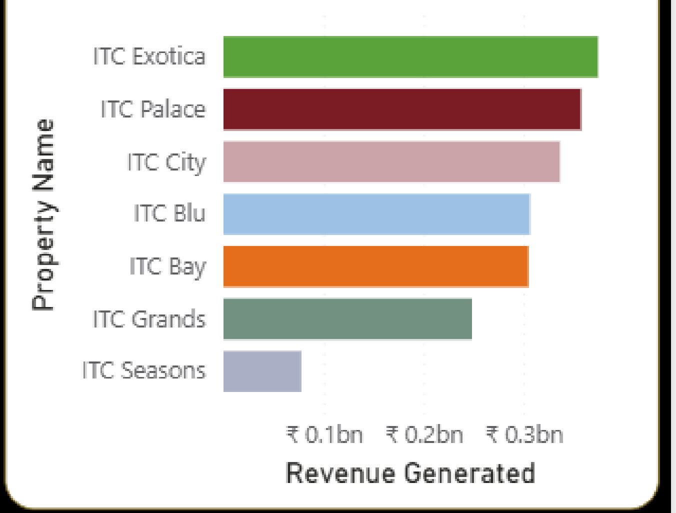
WOW Growth Revenue by Weeknum



Mom Growth by Month



Revenue generated by Property Name



Key Insights - Financial

1. Revenue Performance Overview

- **Total Revenue & Cumulative Revenue** both stand at **₹2.01bn**, showing consistent reporting and strong financial output.
- The **Month-over-Month growth** of **50.29%** reflects a significant revenue recovery or surge, likely driven by seasonal demand or improved occupancy.
- **WoW Growth (1395%)** suggests a sharp increase in week-over-week performance — possibly due to promotional campaigns, events, or improved booking trends.

2. Property-Level Performance

- **Top Performing Property:** *ITC Exotica* contributes the highest revenue among all listed properties.
- **Moderate Performers:** *ITC Palace* and *ITC City* show mid-tier revenue contributions.
- **Low Performers:** *ITC Seasons* and *ITC Grands* exhibit relatively lower revenue levels, signalling a need for marketing or operational optimization.

3. Room Class Contribution

- Premium and Elite categories generate **the largest revenue shares**, implying strong demand for higher-value rooms.
- **Standard rooms** contribute the least, possibly due to pricing or low occupancy rates — a potential area for targeted discounting or bundling strategies.

4. Growth Trends (MoM & WoW)

- **WoW Growth by Week:** Most weeks show stable performance with **sharp peaks**, implying event-driven spikes.
- **MoM Growth by Month:** June shows a temporary dip, followed by a strong recovery in July — reflecting possible seasonal variation or marketing effectiveness.

5. Operational Efficiency

- **ADR (Average Daily Rate)** and **RevPAR (Revenue per Available Room)** follow a stable trend pattern, showing strong revenue consistency with seasonal fluctuations.
- The **steady peaks in ADR/RevPAR** suggest consistent pricing policies and healthy occupancy cycles.

Executive Summary & Strategic Recommendations

Executive Summary

- **ITC Hotels** has achieved a **total revenue generation of ₹2.01 billion**, showing robust financial performance across multiple properties.
- The **Month-over-Month (MoM) Growth stands at 50.29%**, reflecting a strong recovery and increasing demand post-previous months.
- The **Average Daily Rate (ADR)** of **₹14.92K** and **Revenue per Available Room (RevPAR)** of **₹7.35K** indicate solid pricing and occupancy efficiency.
- The **Cumulative Revenue** matches the total generated revenue, suggesting steady accumulation without anomalies.
- Weekly and monthly revenue trends reveal cyclical booking patterns and opportunities for optimization during specific timeframes.

Strategic Recommendations

Optimize Underperforming Properties

- Conduct property-wise diagnostic analysis (especially for ITC Seasons and Grands) to identify causes of low revenue — such as low occupancy, pricing, or market exposure.
- Introduce **geo-targeted promotions** and dynamic pricing to drive bookings in weaker markets.

Strengthen Premium Room Strategy

- Continue leveraging **Premium and Elite categories**, which are driving the highest share of revenue.
- Introduce **value-added packages** (spa, dining, airport transfers) to sustain high ADR without discounting.

Balance Seasonal Demand

- Since June shows a MoM revenue dip, plan **pre-season marketing campaigns** and **off-peak offers**.
- Monitor week-level fluctuations using booking lead time and occupancy correlation to stabilize WoW growth.

Data-Driven Yield Management

- Use **dynamic pricing models** to adjust ADR based on demand elasticity and occupancy rates.
- Focus on **RevPAR optimization** across properties to ensure consistent profitability, not just revenue growth.

Slicers

Finance

Room Occupany

Room Performance

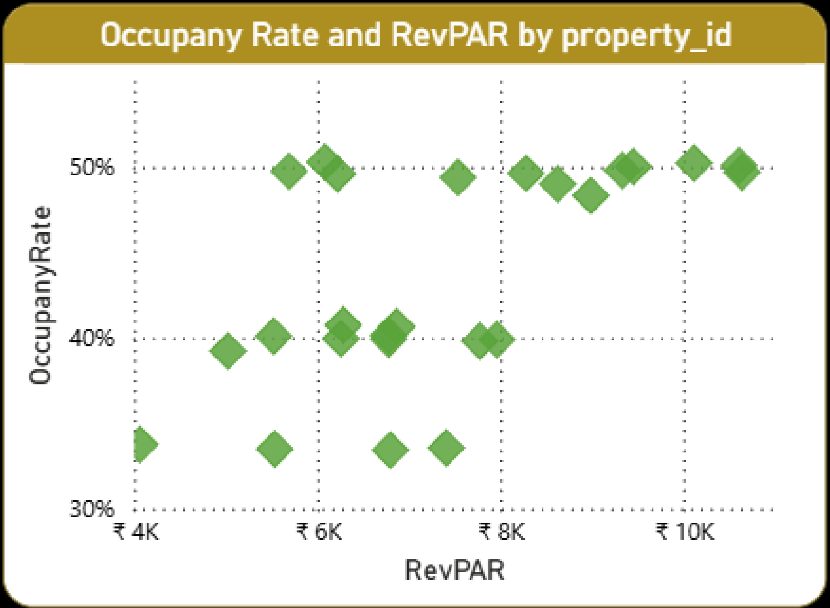
Cancellation

Occupancy & Capacity Analysis

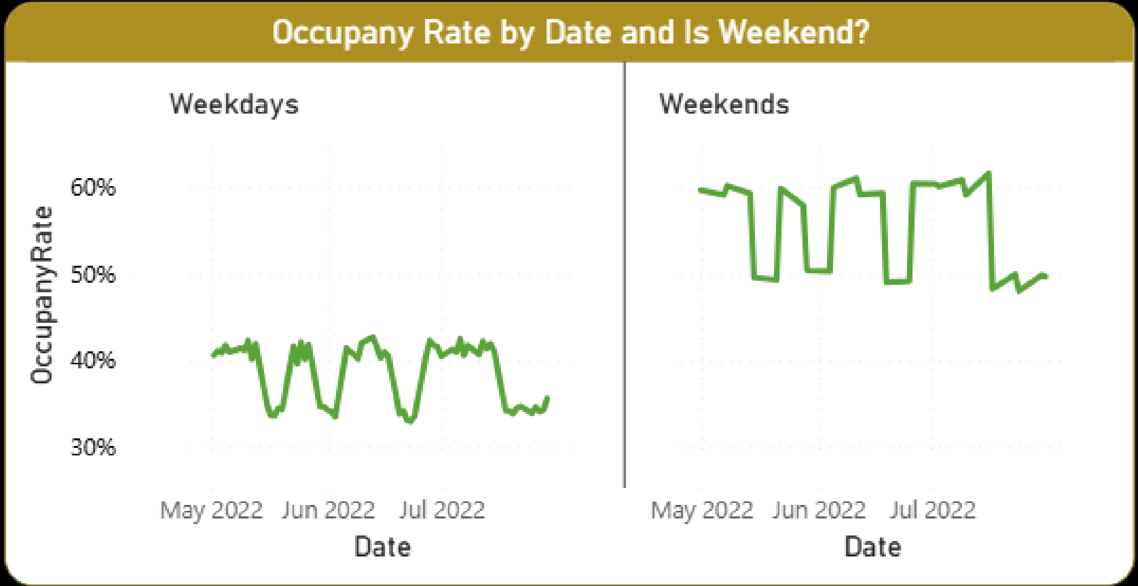
43.5%
Occupany Rate

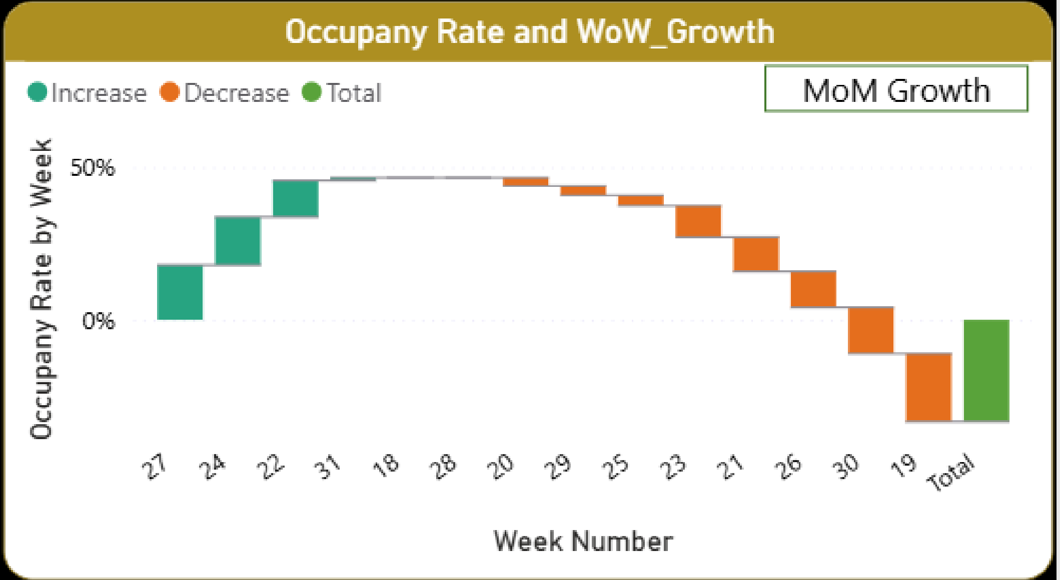
13.14%
Wow Growth

-0.12%
Mom Growth



city	Elite	Premium	Presidential
Agra	50.1%	49.7%	51.6%
Bangalore	39.5%	41.8%	44.7%
Delhi	44.9%	45.6%	47.4%
Goa	50.0%	50.2%	49.5%
Hyderabad	40.7%	41.0%	41.0%
Mumbai	41.5%	39.6%	42.8%
Pune	43.9%	42.5%	45.3%





Key Insights - Occupancy

1 Overall Occupancy and Growth

- **43.5% average occupancy rate** indicates moderate room utilization, below the ideal benchmark (60–70% for premium hotels).
- **WoW Growth** of **+13.14%** signals an improving short-term trend, possibly due to promotions or weekend demand.
- **MoM Growth** of **–0.12%** suggests a minor dip, indicating some stagnation in monthly performance, likely due to off-peak seasonality.

2. City-wise Occupancy Patterns

- **Highest Performers: -** *Agra* (≈50%) and *Goa* (≈50%) show the strongest occupancy across all room types, driven by high tourist appeal and leisure bookings.
- **Moderate Performers: -** *Delhi* maintains occupancy around 45–47%, consistent with its corporate and government travel base.
- **Lowest Performers: -** *Hyderabad (≈41%)* and *Mumbai (≈40–43%)* show underperformance — possibly due to competitive supply or weaker demand recovery.

3. Room Class Trends

- Across cities, **Elite and Presidential rooms** have slightly higher occupancy than Premium rooms, indicating continued demand for luxury experiences.
- **Premium category** shows mild underperformance in some markets, possibly requiring promotional support.

4. Occupancy vs. RevPAR Relationship

- The scatter plot shows a **positive correlation** between **RevPAR (₹4K–₹10K)** and **Occupancy Rate (30–50%)**, confirming that revenue growth is linked to room utilization efficiency.
- Properties achieving **RevPAR above ₹8K** maintain **occupancy near 50%**, suggesting efficient pricing and demand management.

5. Weekday vs. Weekend Dynamics

- **Weekend occupancy consistently outperforms weekdays**, indicating stronger **leisure travel** demand.
- **Weekday occupancy dips to around 35–40%**, reflecting **lower corporate travel** or weak weekday bookings.

6. WoW Occupancy Growth Trend

- Weekly data shows alternating periods of **increase and decline**, with a visible dip during mid-range weeks (weeks 22–26).
- The final weeks show partial recovery, suggesting potential stabilization in demand.0

Executive Summary & Strategic Recommendations

Executive Summary

- The overall **Occupancy Rate stands at 43.50%**, indicating moderate room utilization across ITC Hotels' portfolio.
- **Week-on-Week (WoW) Growth** shows a **positive 13.14%**, signaling recovery momentum, while **Month-on-Month (MoM) Growth** is slightly negative (-0.12%), reflecting short-term stagnation.
- **Weekend occupancy** consistently outperforms weekdays, highlighting strong demand during leisure periods.
- **Top-performing cities** include **Goa (49.68%)**, **Agra (49.83%)**, and **Delhi (45.37%)**, while **Hyderabad (40.61%)** and **Mumbai (41.44%)** show below-average performance.
- The **positive correlation between Occupancy Rate and RevPAR** (up to ₹10K at 50% occupancy) indicates that capacity utilization directly drives revenue efficiency.

Strategic Recommendations

Strengthen Weekday Occupancy

- Launch **weekday corporate offers** and **business traveler packages** in cities like *Hyderabad, Pune, and Mumbai*.
- Partner with **corporate accounts and travel aggregators** to fill weekday gaps.

Leverage High-Performing Markets

- In *Agra* and *Goa*, capitalize on the strong demand by introducing **premium upselling strategies** (room upgrades, bundled services).
- Deploy **dynamic pricing** to maximize ADR during weekends and high-traffic period.

Improve Underperforming Cities

- For *Hyderabad and Mumbai*, conduct **root-cause analysis**—focus on competitor pricing, marketing reach, and local events.
- Introduce **targeted discount campaigns** or **event partnerships** to boost visibility

Enhance Occupancy and Revenue Synergy

- Use **RevPAR and Occupancy trend analysis** to identify the optimal pricing threshold for each city and room type
- Focus on **balancing volume and rate** — avoid deep discounts that may hurt RevPAR.

Focus on Seasonal and Event-Based Campaigns

- Since weekends perform better, roll out **short-stay or weekend getaway packages**.
- Use **seasonal demand forecasting** to allocate marketing budgets effectively (especially for off-peak months).

Slicers

Finance

Room Occupany

Room Performance

Cancellation

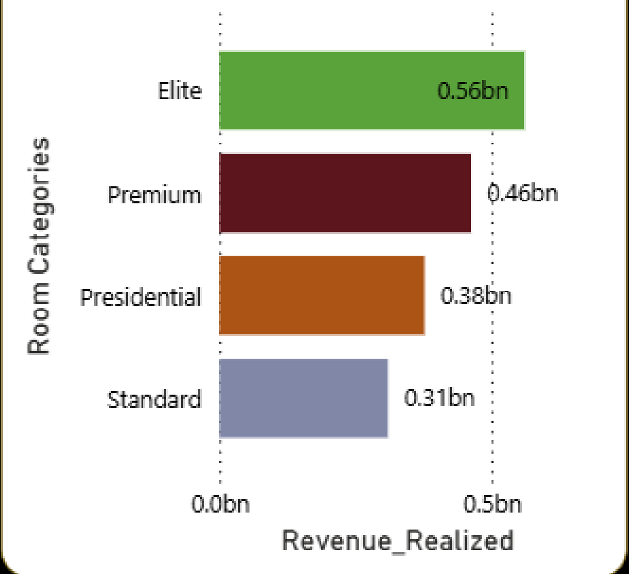
Room Category Performance & Booking Insights

3.71
Avg of booking_lead_time

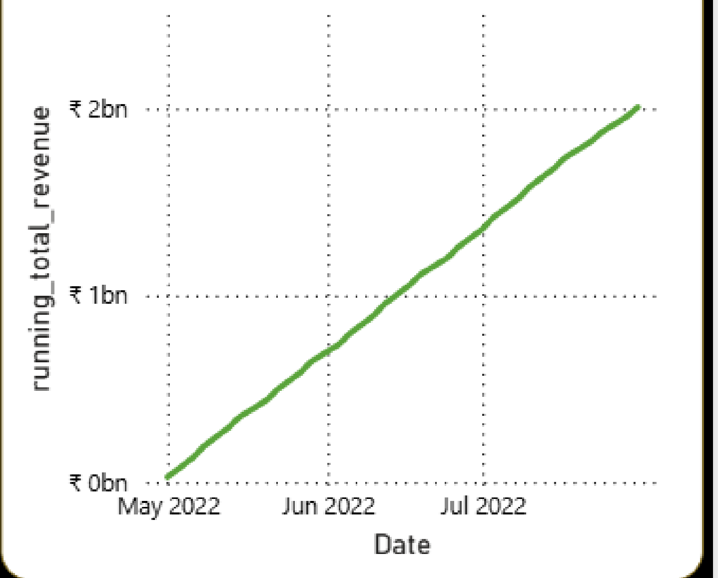
2.37
Average Length of Stay

property_name	Elite	Premium	Presidential	Standard
ITC Bay	2.75	2.73	2.73	2.75
ITC Blu	2.75	2.80	2.69	2.73
ITC City	1.76	1.74	1.77	1.79
ITC Exotica	2.74	2.74	2.76	2.74
ITC Grands	2.75	2.73	2.75	2.70
ITC Palace	1.77	1.79	1.75	1.78

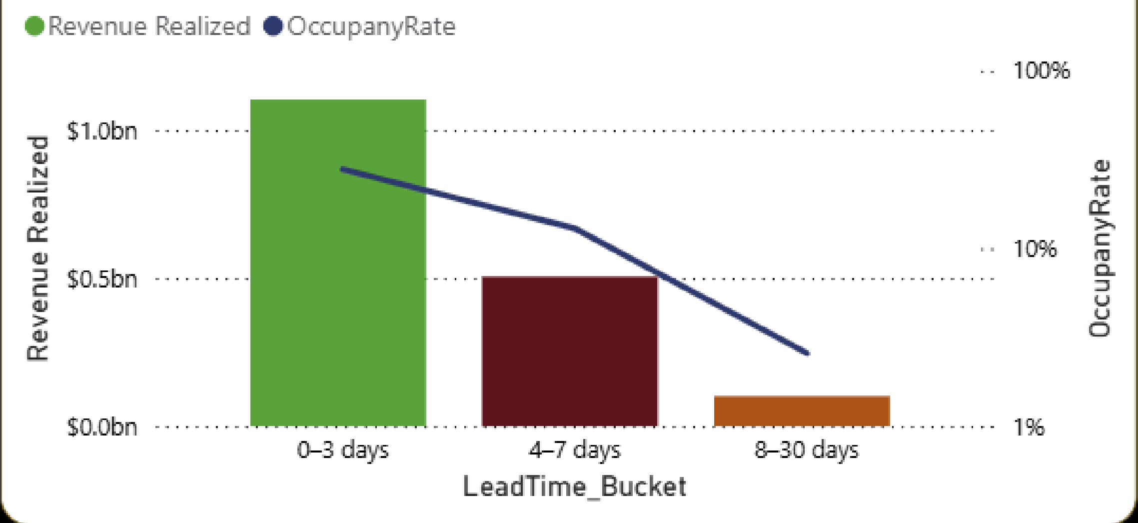
Revenue_Realized by Room_Class



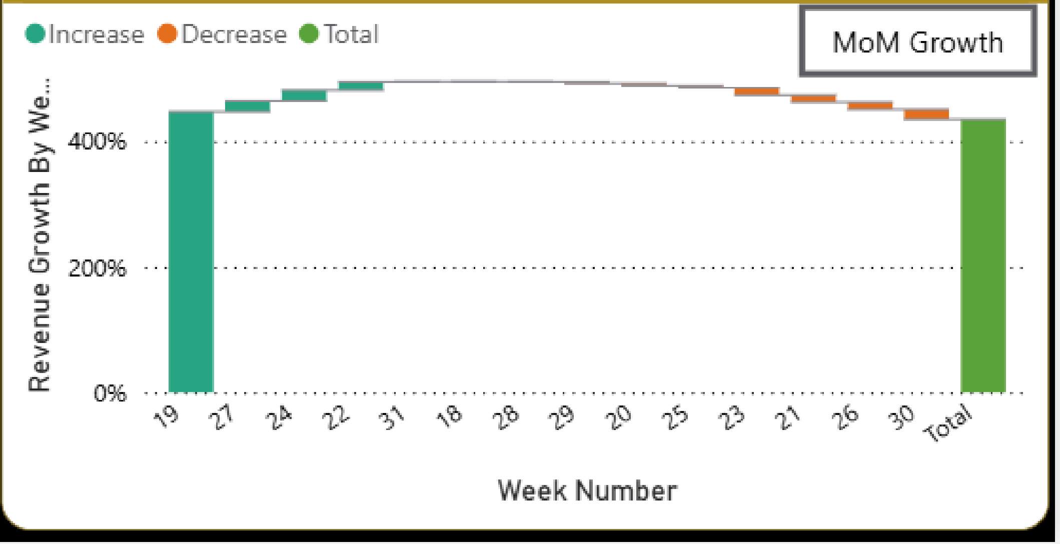
running_total_revenue by Date



Revenue Realized and OccupanyRate by LeadTime_Bucket



Revenue by Week



Key Insights – Room Performance

1. Booking Behaviour & Patterns

- **Average Booking Lead Time:** 3.71 days → Guests tend to book **just a few days before check-in**, indicating spontaneous travel or short booking windows.
- **Average Length of Stay:** 2.37 days → Most stays are short-term, reflecting **business or short leisure trips** rather than long vacations.

2. Revenue Performance by Room Class

- **Elite rooms lead revenue generation (₹0.56bn)**, followed by **Premium (₹0.46bn)** and **Presidential (₹0.38bn)** categories.
- **Standard rooms (₹0.31bn)** contribute the least, possibly due to lower pricing or limited occupancy.
- Higher-tier categories (Elite & Premium) dominate total revenue share, showing strong demand for **luxury and comfort-focused segments**.

3. Lead Time Impact on Revenue and Occupancy

- Bookings made **within 0–3 days** before check-in generate **the highest revenue (~₹1bn)** and maintain the highest occupancy rate.
- As lead time increases (4–7 days, 8–30 days), both **revenue and occupancy drop significantly**, suggesting limited success in attracting early bookers.
- This emphasizes a **high dependence on last-minute bookings**, which can expose the business to volatility if demand softens.

4. Revenue Trend Over Time

- The **running total revenue** shows a **steady, linear upward trajectory**, indicating consistent growth without major drops — a sign of **sustained demand** and strong operational stability.
- The **weekly revenue growth chart** shows fluctuations with alternating **increase/decrease periods**, but overall, the trend closes on a **positive total growth**, reinforcing resilience in performance.

5. Property-Level ALOS Patterns

- Across all ITC properties, **Elite and Premium categories** have slightly higher ALOS (2.7–2.8 days), while **Presidential and Standard rooms** show shorter stays (1.7–1.8 days).
- This suggests that guests choosing **higher-end rooms stay marginally longer**, likely due to the perceived value and comfort of those segments.

Executive Summary & Strategic Recommendations

Executive Summary

- **ITC Hotels** exhibits a healthy **average booking lead time of 3.71 days** and an **average length of stay (ALOS) of 2.37 days**, reflecting short-stay, near-term booking behavior common in urban and leisure segments.
- **Elite rooms** generate the **highest revenue share (32.79%)**, followed by **Premium (27.05%)**, showing strong preference for higher-tier room categories.
- The **MoM Revenue Growth** indicates a **decline in June (-5%)**, followed by stabilization in July, pointing to a temporary mid-season slowdown.
- **Revenue realization and occupancy** decline as the booking lead time increases, indicating that most bookings (and revenue) come from **last-minute stays (0–3 days)**.
- **Running total revenue** demonstrates a **steady upward trend**, signifying consistent accumulation of income and strong ongoing performance..

Strategic Recommendations

Optimize Advance Bookings

- Launch **early-bird incentives** (e.g., 10% off or added benefits for 7+ day advance bookings) to **extend the booking window** and reduce last-minute dependency.
- Collaborate with **online travel agencies (OTAs)** to promote "Book Early & Save" packages

Strengthen Mid-Tier Room Performance

- Introduce **value-based bundles** for *Premium and Standard* categories (e.g., meal packages, airport pickup) to lift revenue and occupancy.
- Review **pricing elasticity** to ensure competitive positioning relative to the Elite tier.

Leverage High-Performing Segments

- Continue promoting **Elite and Premium categories** as core revenue drivers through exclusive loyalty rewards, upselling campaigns, and personalized offers for repeat guests.
- Focus marketing on **short-stay premium travellers** (corporate and couples).

Extend Guest Stay Duration

- Introduce offers like **"Stay 3 Pay 2"** or **"Weekend Extension Deals"** to encourage longer stays and improve ALOS.
- Promote **cross-property stay programs**, where guests can extend their stay at a partner ITC hotel nearby at discounted rates.

Maintain Revenue Growth Momentum

- Keep monitoring **running total and week-over-week revenue trends** for early signs of plateauing.
- Utilize **dynamic pricing models** to sustain upward growth during low-demand weeks.

Slicers

Finance

Room Occupany

Room Performance

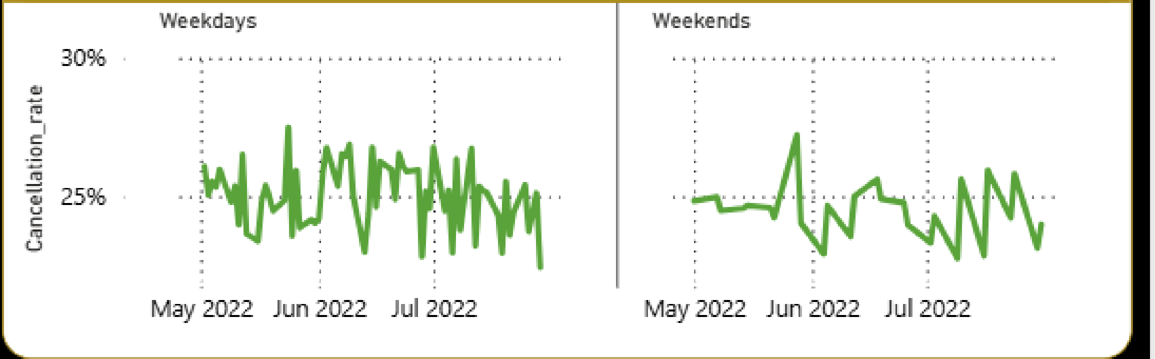
Cancelletion

Cancellations & Lost Revenue Analysis

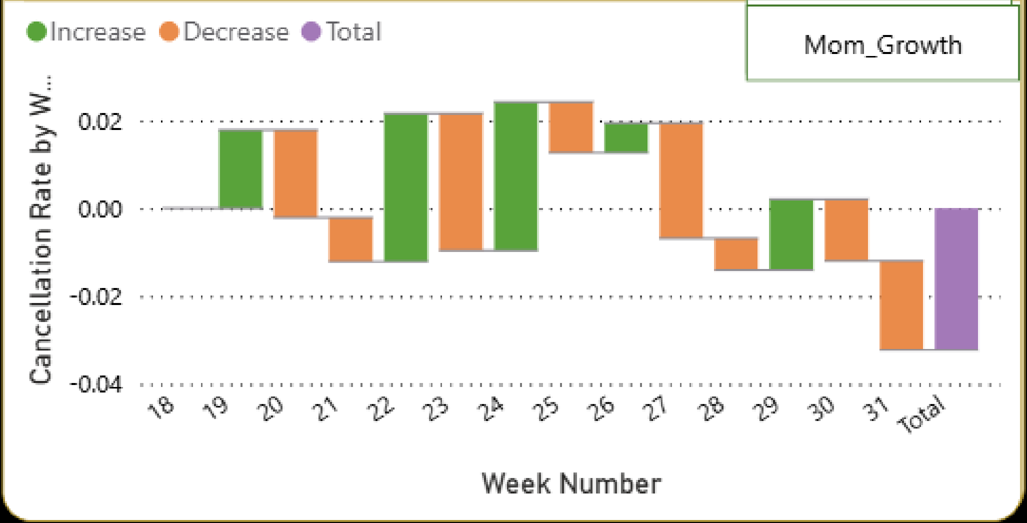
1.45%	33.42K	₹ 497.96M	24.83%	-0.01
wow_growth	Cancelled_bookings	Lost Revenue (Cancellati...	Cancellation_rate	Mom_Growth

property_name	Elite	Premium	Presidential	Standard	Total
ITC Palace	25.58%	24.20%	25.49%	25.20%	25
ITC Grands	25.45%	25.10%	23.09%	25.31%	25
ITC City	24.87%	25.78%	24.99%	24.35%	24
ITC Bay	25.32%	25.45%	22.50%	24.79%	24
ITC Seasons	24.61%	26.20%	23.51%	24.42%	24
ITC Blu	24.10%	24.64%	27.31%	24.65%	24
Total	**24.96%**	**24.88%**	**24.44%**	**24.79%**	**24.**

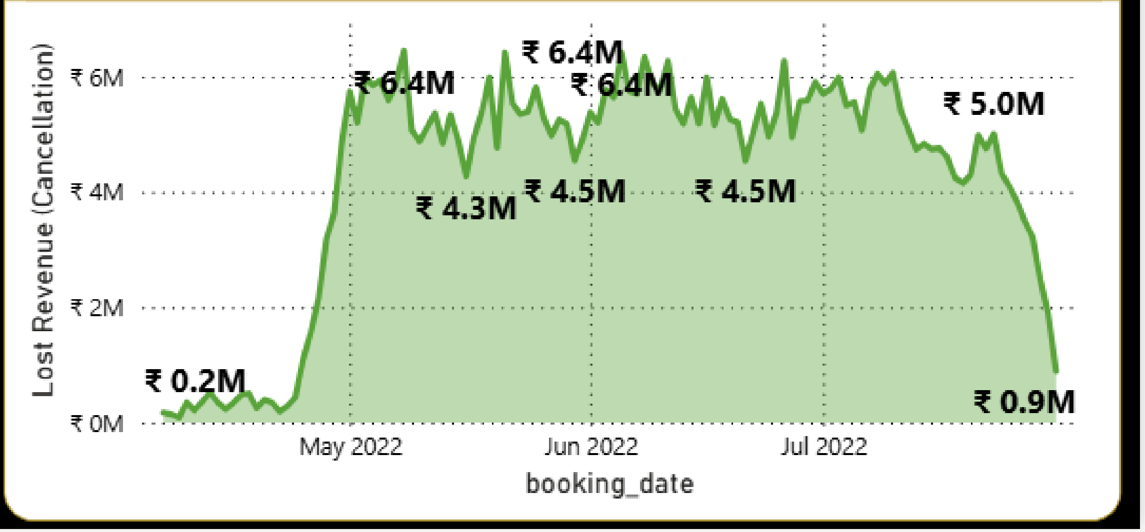
Cancellation_rate by Date and Is Weekend?



Cancellation Rate by Week by Week Number



Lost Revenue (Cancellation) by booking_date



Key Insights – Cancellation

1. Overall Cancellation Performance

- **Total Cancelled Bookings:** 33.42K
- **Cancellation Rate:** 24.83% — roughly 1 in 4 bookings are cancelled.
- **Lost Revenue:** ₹497.96M (₹0.5B), indicating a **significant impact on revenue realization**.
- **Week-over-Week (WoW) Growth:** +1.45%, showing **minor short-term rise** in cancellations.
- **Month-over-Month (MoM) Growth:** -0.01, implying **stability** after prior fluctuations.

2. Property-Wise Cancellation Insights

Property	Avg Cancellation Rate	Notable Patterns
ITC Palace	25.8%	Slightly above average — possibly due to high-priced or corporate bookings.
ITC Grands	25.4%	Consistent but elevated; may indicate pricing or booking policy issues.
ITC City	24.9%	Balanced performance, in line with average.
ITC Bay	25.3%	Slightly high but stable.
ITC Seasons	24.6%	Lower end of range, indicating relatively better booking retention.
ITC Blu	24.1%	Lowest cancellation rate — could be a benchmark for best practices.

Insight: Variance across properties is **minimal (±1%)**, implying **systemic cancellation behavior** rather than location-specific issues

3. Cancellation by Room Category

- **Presidential rooms (27.3%)** and **Standard rooms (25.3%)** show **higher cancellation tendencies**, possibly due to:
  - Flexible booking policies.
  - Price sensitivity or alternate booking options.
- **Elite and Premium rooms** have slightly lower cancellation rates (~24.5%), suggesting stronger commitment among higher-paying guest.

4. Temporal Trends

- **Weekday vs Weekend:**
  - **Higher weekday cancellation rates**, likely due to corporate or business travel adjustments.
  - **Lower weekend cancellations**, consistent with confirmed leisure stays.
- **Weekly Analysis:**
  - Alternating periods of increase and decrease, showing **seasonal or event-driven fluctuations**.
- Recent weeks indicate a **downward momentum**, suggesting improved control or demand stability.

5. Lost Revenue from Cancellations

- **Peaks of ₹6.4M** observed around May–June 2022, aligning with potential **seasonal or promotional booking surges** followed by withdrawals.
- Average lost revenue post-peak stabilizes between **₹4.3M–₹5M**, indicating consistent but **avoidable leakage**.
- Recent decline (₹0.9M) reflects **possible improvement in retention or fewer cancellations**.

Executive Summary & Strategic Recommendations

Executive Summary

- **ITC Hotels** experienced a **24.83% overall cancellation rate**, resulting in an estimated **₹199.18 million in lost revenue** due to booking cancellations.
- A total of **33.42K bookings were cancelled**, indicating a substantial volume impact on realized revenue.
- The **Week-on-Week (WoW) cancellation growth** shows mixed patterns with frequent fluctuations, suggesting irregular booking behaviors or external factors (events, pricing, travel restrictions).
- **Month-on-Month (MoM) growth in revenue (-0.01)** remains nearly flat, implying that cancellation recovery efforts are not yet translating into financial improvements.
- **Weekday cancellations** are consistently higher than weekends, possibly due to business travel dependencies or flexible corporate booking policies.

Strategic Recommendations

Strengthen Cancellation Policies

- Introduce **tiered cancellation fees** — e.g., free cancellation up to 3 days before check-in, partial refund within 48 hours, and no refund within 24 hours.
- Provide **non-refundable but discounted booking options** to encourage firm commitments

Analyse and Target High-Cancellation Segments

- Perform **segmentation by source (OTA, corporate, direct)** to identify channels with the highest cancellation ratios.
- For **Presidential and Standard rooms**, review booking flexibility, payment terms, and guest demographics to design **targeted retention strategies**

Introduce Incentives for Retention

- Offer **credit vouchers** or **future stay discounts** instead of full refunds for cancellations.
- Implement **“Hold My Booking”** or **“Change My Dates”** options to reduce outright cancellations

Monitor and Predict Cancellations

- Develop a **predictive model** in Power BI using historical data (lead time, booking source, room type, stay length) to **forecast potential cancellations** early.
- Send **automated pre-stay confirmations or reminders** to minimize no-shows and last-minute cancellations.

Monitor and Predict Cancellations

- Develop a **predictive model** in Power BI using historical data (lead time, booking source, room type, stay length) to **forecast potential cancellations** early.

Leverage Insights from Low-Cancellation Properties

- Study operational or customer experience practices at **ITC Blu** (lowest cancellation rate) and **ITC Seasons** to replicate effective policies across other hotels.

Strategic Takeaways & Conclusion

Strategic Takeaways:

- **Occupancy & Growth:**
  Overall occupancy stands at ~43.5%, with **positive WoW growth (+13%)** but a **slight MoM decline**, indicating short-term improvement in demand yet requiring sustained marketing efforts.
- **Revenue Drivers:**
  Elite and Premium rooms contribute the highest realized revenue, while **short lead-time bookings (0–3 days)** generate the majority of revenue — highlighting a **high dependency on last-minute bookings**.
- **Lead Time & Stay Pattern:**
  Average booking lead time is **3.7 days**, and average stay length is **2.4 days**, suggesting a predominantly **short-stay, quick-decision traveler profile**.
- **Cancellations & Lost Revenue:**
  Cancellations remain high at **~25%**, causing a **revenue loss of nearly ₹498M**.
  Weekday cancellations are more frequent, likely due to corporate travel adjustments.
- **Property & Segment Insights:**
  ITC Blu and ITC Seasons show better booking retention, while **Presidential and Standard categories** face higher cancellations — potential focus areas for tightening policy or incentives.

Conclusion

- ITC Hotels demonstrates **strong short-term growth** and **steady revenue realization**, yet faces **significant leakage through cancellations** and **reliance on short-lead bookings**.
  To enhance performance, the chain should **balance demand through early booking campaigns**, **introduce stricter yet flexible cancellation terms**, and **leverage predictive analytics** for better booking retention and revenue optimization..